ANALYSIS OF MORTGAGE LOAN APPLICATIONS (IOWA 2024)

The Outliers

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MORTGAGE LOAN APPLICATION OUTCOMES - 2024

- Exploratory analysis
- 3 Main Questions:
  - Is there a relationship between the applicant's sex and the action taken?
  - Is the average loan amount different for insured and uninsured loans among disbursed loans?
  - Is there a relationship between income and loan amount among approved loans?

THE DATA

- Mortgage application data for the state of Iowa for the year 2024
- Obtained from FFIEC Home Mortgage Disclosure Act (HMDA) website
  - https://ffiec.cfpb.gov/data-browser/data/2024?category=states
  - All data comes from sources available in public records
- The dataset contains 110,208 applications, with 99 variables

DATA CLEANING

- Focused on 9 variables
  - Race, Ethnicity, Sex, Dwelling Type, Loan Status, Loan Type, Loan Purpose, Loan Amount, Income
- Excluded incomplete or withdrawn loan applications
  - 12,138 application withdrawn
  - 3,411 closed for incompleteness
- During the analysis appropriate transformations like recoding, mutating, filtering, regrouping, etc., are performed to create relevant data frames

RESEARCH QUESTION 1: APPLICANT'S SEX V. ACTION TAKEN

- Removed all data points where sex was not available
  - 75543 observations analyzed
- Applicant's Sex divided into 3 groups
  - Male, Female, Joint
- Divided loan status into two groups
  - Approved loans
    - Combined "Loan Originated" and "Application approved but not accepted"
  - Denied loans

Q1 METHODOLOGY

- Among Iowa loan applicants, is there a difference in loan approval rates between the Sex of the loan applicant?
- The distribution of Loans by gender are shown in Figure 1a.

Fig 1a. Distribution of Loan Status by Sex

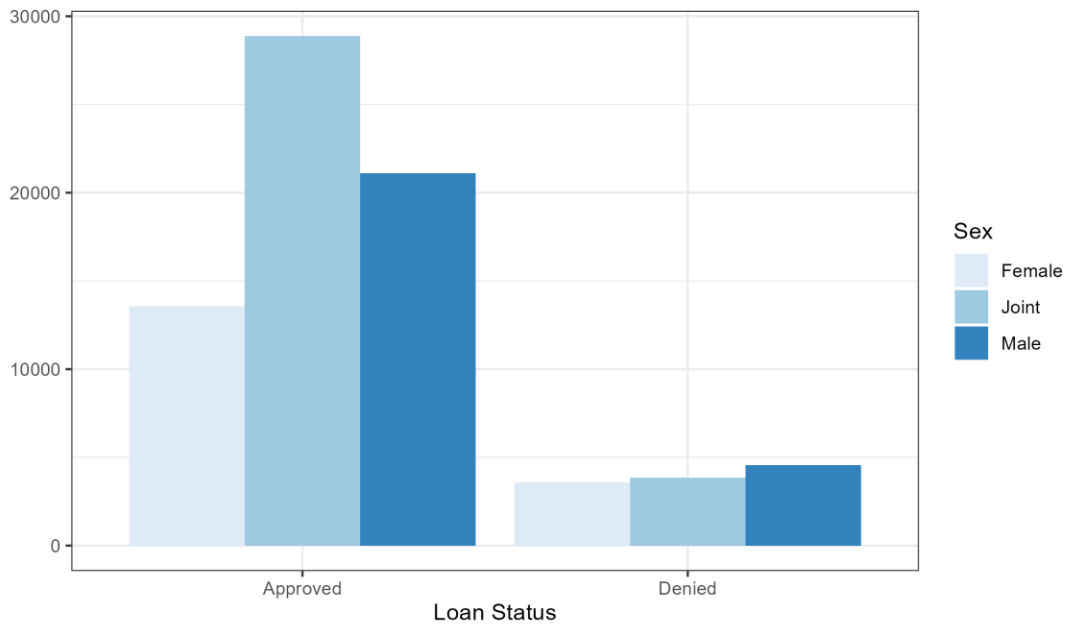


Table 1: Contingency Table

	Male	Female	Joint
Approved	21109	13568	28870
Denied	4576	3584	3836

Table 2: Proportion of Approvals and Denials by Sex

	Male	Female	Joint
Approved	82.2	79.1	88.3
Denied	17.8	20.9	11.7

Q1: RESULTS

- Performed a Chi-Square test of independence to determine if loan approval rates are independent of applicant sex
  - $p<0.0001$
  - Loan applications from females and joint applicants contribute most (Table3)

Table 3: Squared Residuals from Test of Independence

	Male	Female	Joint
Approved	11.446	51.298	66.992
Denied	60.634	271.742	354.881

- Result indicate that among Iowa loan applicants in 2024 there is significant evidence that loan approval or denial is not independent of the applicant's sex

RESEARCH QUESTION 2

- Do Insured and Uninsured Loans Differ in Loan Amount?

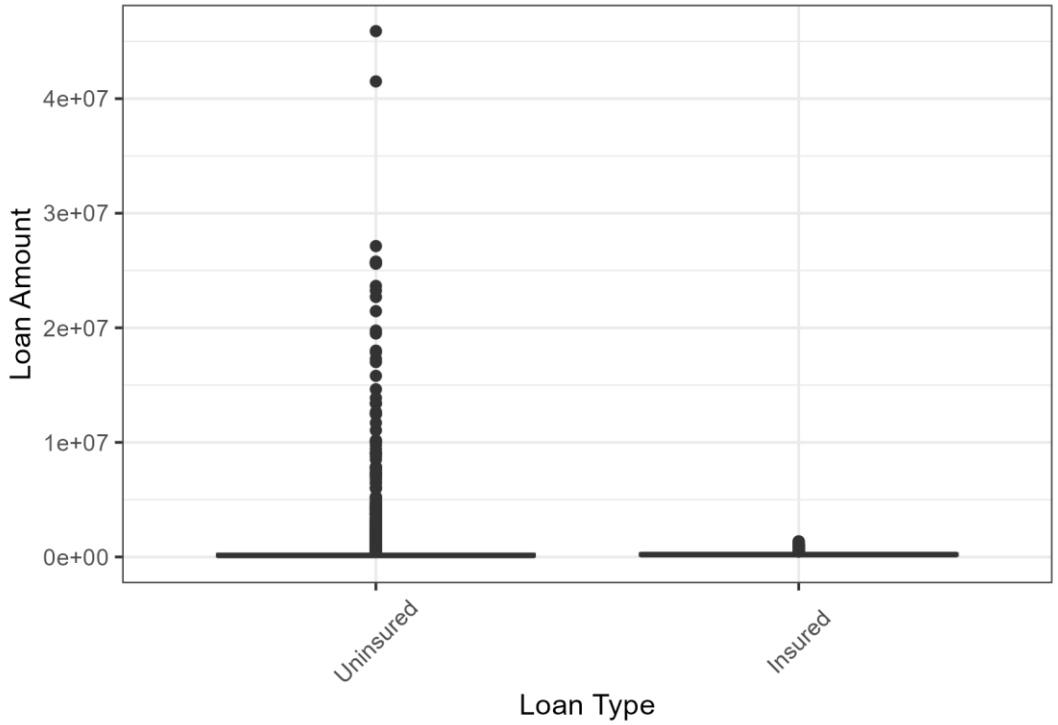
  - Analysis restricted to originated + purchased (disbursed) loans
  - Insured loans: Backed by
    Federal Housing Authority (FHA)
    Veteran Affairs (VA)
    Rural Housing Service (RHS)
    Farm Service Agency (FSA)
  - Uninsured loans: Conventional loans
  - Compares mean loan amount between groups (t-test)
  - n = 78,649 observations

Q2: METHODOLOGY

Loan Amounts Are Highly Right-Skewed

- Distribution heavily skewed
- Extreme outliers up to $45M
- Violates assumptions:
  - Normality
  - Equal Variance
- Motivates log transformation

Fig 2a. Distribution of Loan Amount by Type



Q2: METHODOLOGY(CONTD.)

Log Transformation Stabilizes Variance

- Spread becomes more uniform
- Fewer large outliers
- Suitable for two-sample t-test

Assumptions

- Independence: reasonable
- Variances: SD ratio < 2
- Normality: Some deviation but acceptable with large n
- Tested robustness by trimming top 1% of values

Fig 2b.1. Residual Plot for log(Loan Amount)

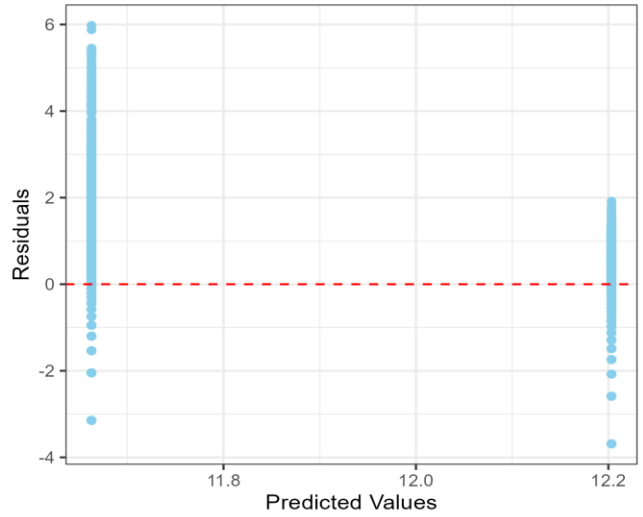
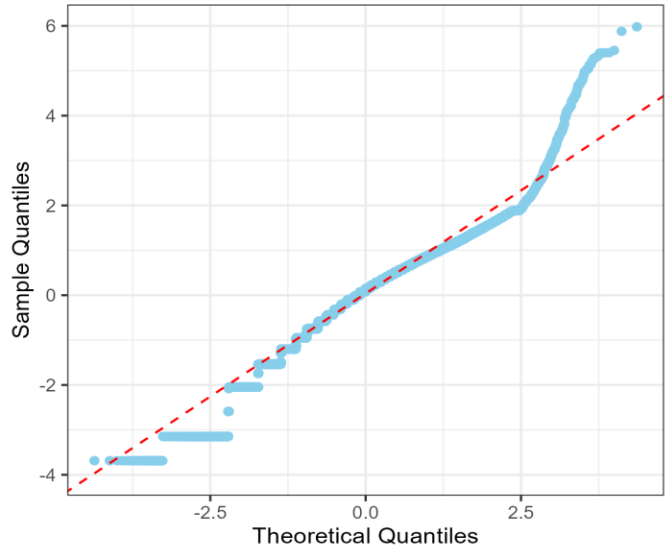


Fig 2b.2. QQ plot for log(Loan Amount)



Q2: RESULTS

Insured Loans Have Higher Loan Amounts

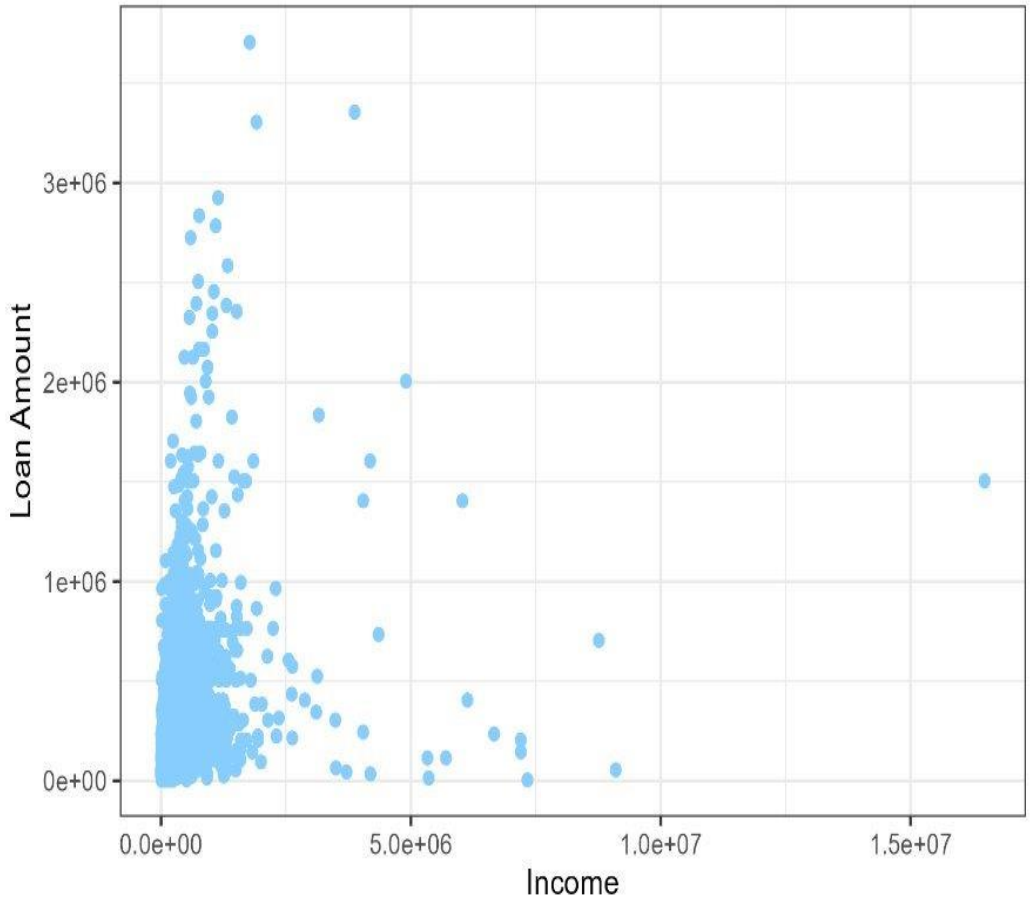
- p-value < 0.0001
  Extremely strong evidence of a difference in average loan amount
- Difference in mean log loan amount = 0.54
- Median loan amount for insured loans is 1.72 times higher than that of uninsured
- 95% CI = (1.70,1.74)

- Tested robustness by trimming top 1% of values
  More stable variance and normality
  Roughly similar results

WHY IS A REGRESSION MODEL NEEDED?

- Loan type explains some variation
- But income also plays a major role
- Regression lets us model multiple predictors

Fig 3a. Loan Amount vs Income



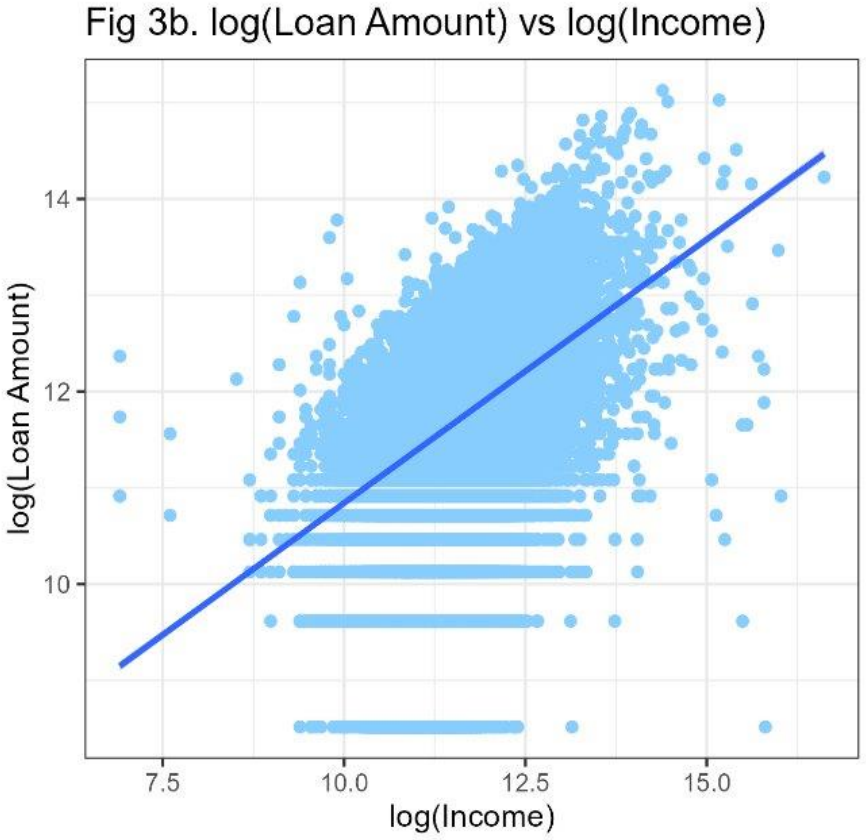
RESEARCH QUESTION 3

- Among approved loans, is there a linear relationship between an applicant's income and loan amount ?
    - Analysis restricted to:
        Originated loans
        Loans approved but not accepted
    - Only applicants with valid income are considered
    - n = 64,851 observations
    - Studies relationship between income and loan amount through regression

Q3: SLR MODELING

Log-Transforming Reveals a Linear Trend

- Relationship is non-linear on raw scale
- Both variables are log transformed
- Linear association appears



Q3: SLR MODELING (CONTD.)

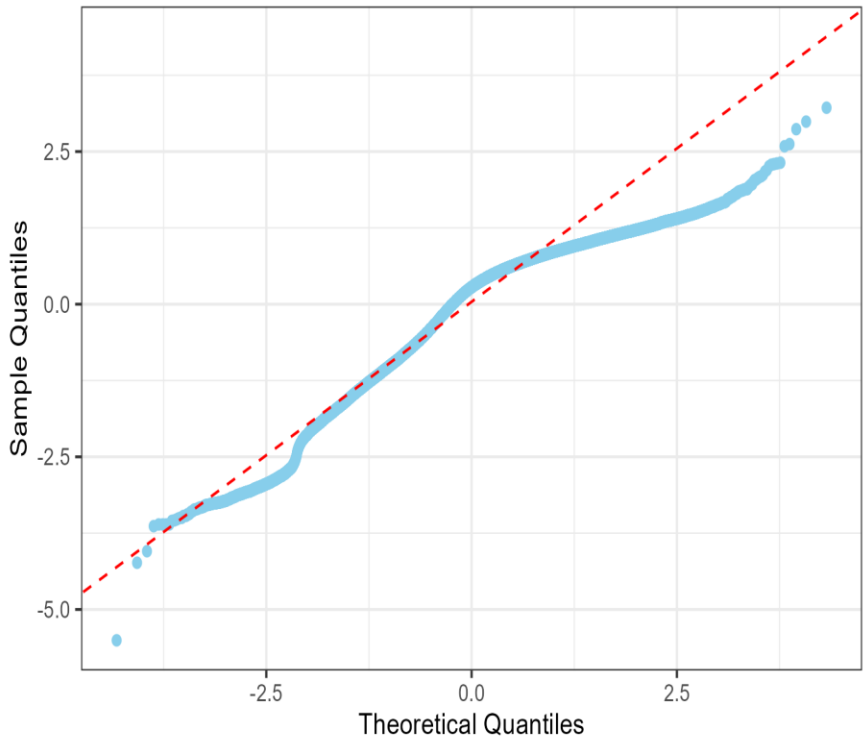
SLR Diagnostics

- Variance roughly constant
- Slight deviation at tails but acceptable
- No major dependence issues

SLR Results

- Estimate of coefficient for income, slope $\hat{\beta}_1 = 0.5423$
- p-value < 0.0001
- A 1% increase in income is associated with approximately 0.5423% increase in loan amount

Fig 3b.2. QQ Plot for log(Loan Amount) vs. log(Income)



Why is MLR Needed?

SLR Is Not Enough: More Predictors Matter

- Other predictor variables like loan type and loan purpose also affect loan amount
- Need a model that accounts for multiple variables
- Leads to **Multiple Linear Regression**

Q3 MLR

- Considered other variables that might be linearly related to log loan amount alongside log income
  - Loan Type (Conventional, FHA insured, VA guaranteed, RHS/FSA guaranteed
  - Loan purpose (Home purchase, home improvement, refinancing, cash out refinancing, other)
  - Sex (Female, Male, Joint)
  - Race (White, African American, Asian, etc)
  - Ethnicity (Hispanic or Latino, Joint, Not Hispanic or Latino)
  - Dwelling type (Multifamily manufactured, multifamily site built, single family manufactured, single family site built)

Q3 MLR (CONT)

- Fit each variable separately with log income to model log loan amount
  - Only the type of loan and purpose of the loan were statistically significant at $\alpha = 0.05$ level
- Using automated model selection with AICc as the criteria; 4 possible models

model	prop_sresid_within2	num_sresid_outside4
logactual_incomexloan_purpose + logactual_incomexloan_type	0.9534245	460
logactual_income + loan_purpose + loan_type	0.9539599	483
logactual_income + loan_purpose	0.9532346	521
logactual_income + loan_type	0.9623355	5

model	num_outside_cook	AIC
logactual_incomexloan_purpose + logactual_incomexloan_type	2134	131653.0
logactual_income + loan_purpose + loan_type	2482	131875.8
logactual_income + loan_purpose	3645	134282.5
logactual_income + loan_type	930	150646.6

Table 4: Model Selection

BEST MLR MODEL

- Final goal is to have a MLR model that is more interpretable and able to fit current data well
  - Possibly be used to predict new data
- Things considered to obtain the best model
  - Required assumptions: errors are independent, have constant variance, normally distributed
  - Also considered: multicollinearity

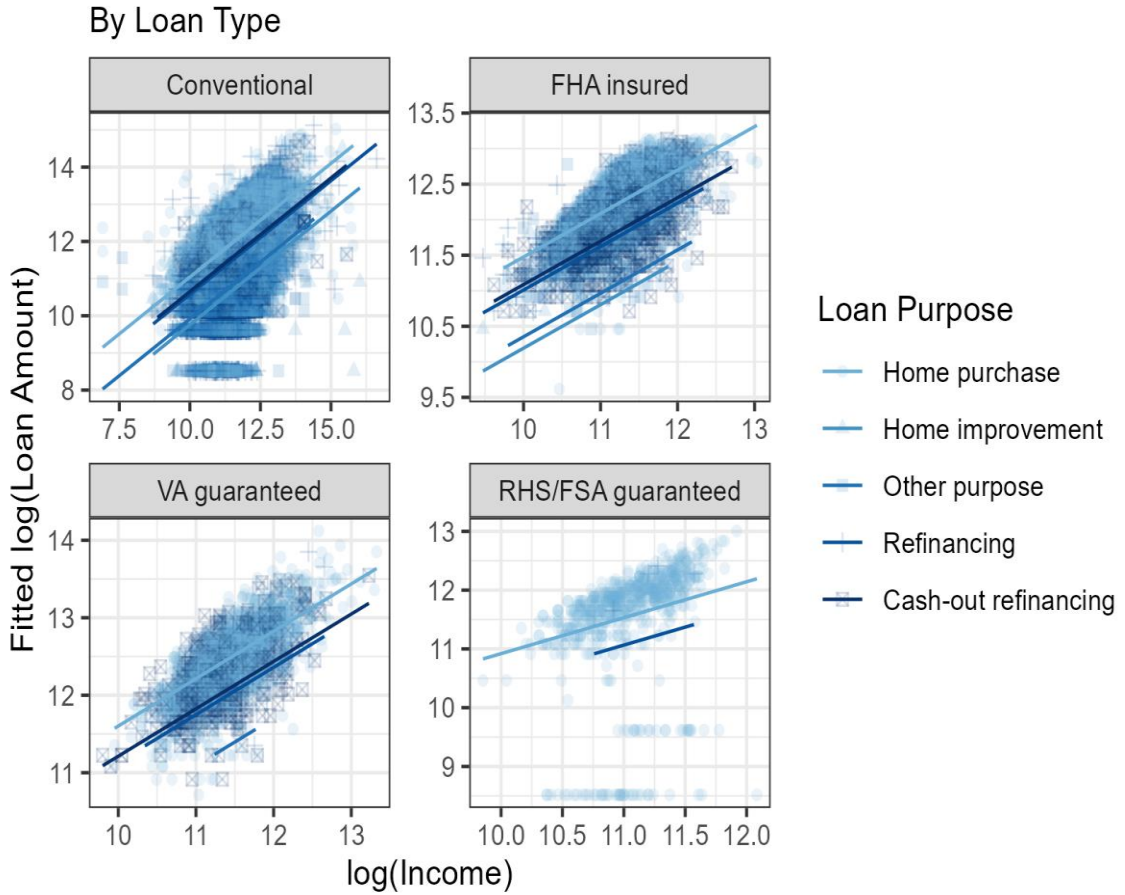
BEST MLR MODEL (CONT.)

- Model 1: Assumptions were fine; possible non-normality
  - Extreme multicollinearity
    - adjusted generalized variance inflation factors (GVIFs) > 18
  - Hard to interpret with 7 different interaction terms
  - AIC difference of 232 for next model
- Model 2: Assumptions were fine; possible non-normality
  - Easier to interpret
- Model 3: Same as model 2
  - Higher AICc
- Model 4: Better normality
  - Highest AICc

BEST MLR MODEL (CONTD.)

- For our goals, model 2 is the best
  - log loan amount ~ log income + type of loan + purpose of loan
  - Easy to interpret:
    - i.e.: for a conventional loan for a home purchase, a one percent increase in income is associated with a 0.611 percent increase in loan amount
- Mean residuals on the original scale
  - Model 2: 74750.17 dollars
  - Model 1: 74616.15 dollars

Fig 3g. Effects of Income and Loan Purpose



MLR COMMENTS

- MLR model only explains 40.9 percent of the variability
  - Removing some observations may increase this
- If we wanted the strongest possible model available for the data for fitting and predicting
  - Could possibly explore ridge regression or lasso

CONCLUSIONS

- Q1
  - Loan outcome is not independent of the applicant's sex
- Q2
  - Insured loans have higher loan amounts
- Q3
  - Best model was log loan amount ~ log income + type of loan + purpose of loan